Identification of target genes that regulate metabolic pathways in cervical cancer from HPV16 and HPV18 positive patients

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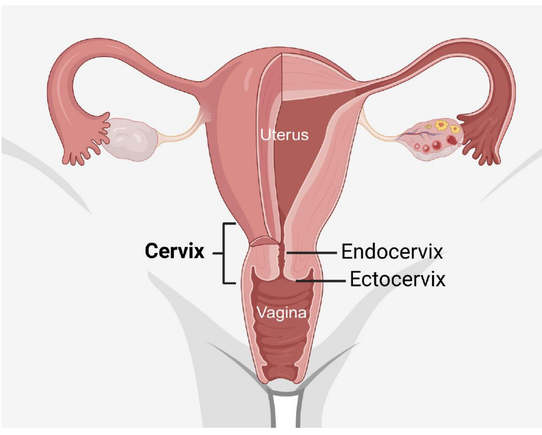
BACKGROUND

Cervical cancer (CC) is a major cause of cancer mortality in women, with over 660,000 new cases reported in 2022 [1]. More than 95% of CC cases are linked to **high-risk Human Papillomavirus (hrHPV)**, mainly subtypes 16 and 18 [2]. CC is primarily classified into **squamous cell carcinoma (SCC)** and **adenocarcinoma (AC)**, each associated with distinct cellular origins and hrHPV subtypes [3,4]. Understanding genetic and pathway differences across CC types, including rare variants, is crucial for developing precise diagnostic and therapeutic strategies.

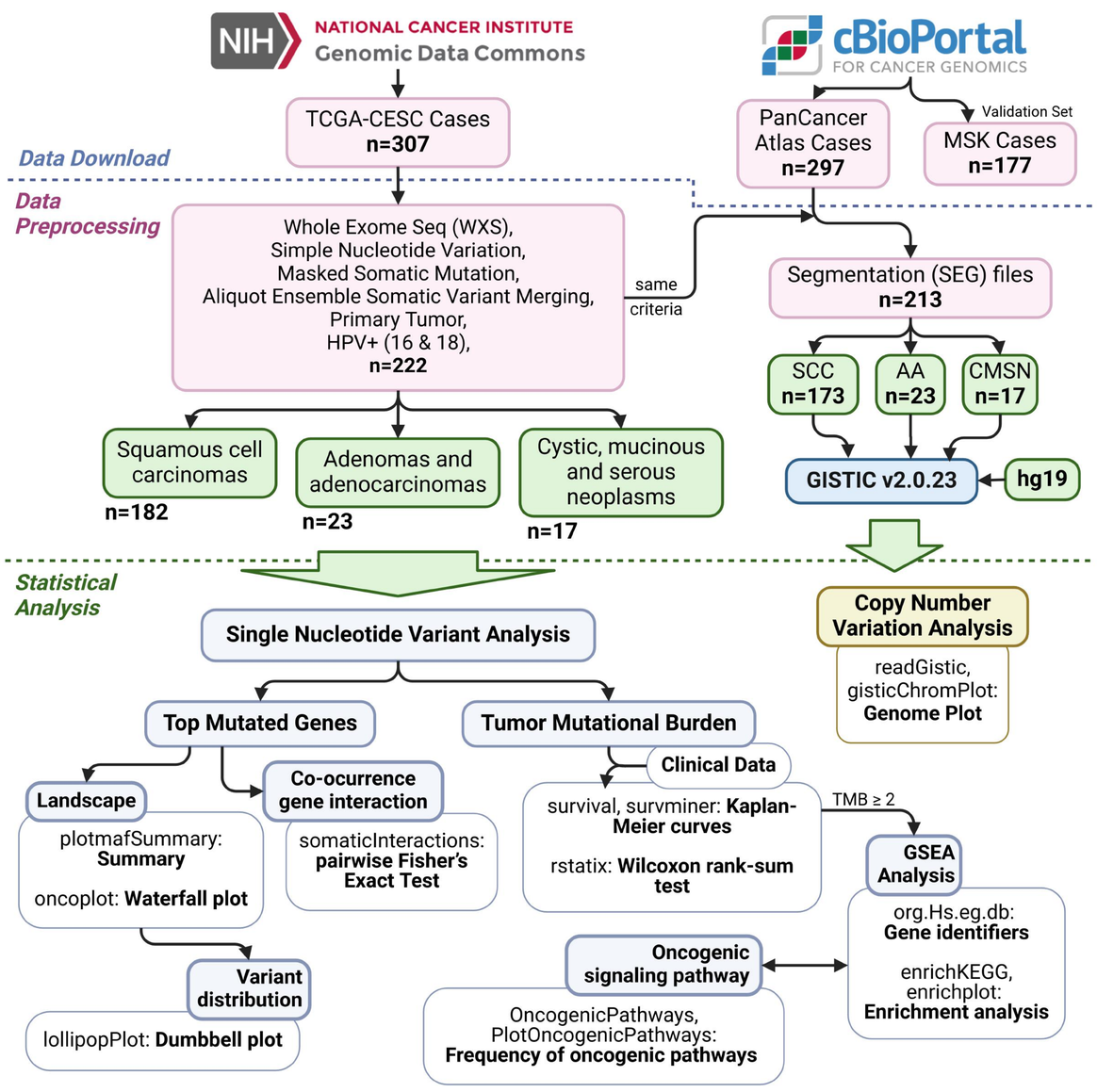
Squamous Cell Carcinoma (SCC) ~80-90%	Adenocarcinoma and Adenocarcinoma(AA) ~10-20%
SCCs develop from squamous cells in the exocervix	AAs develop from glandular cells in the endocervix

OBJECTIVE



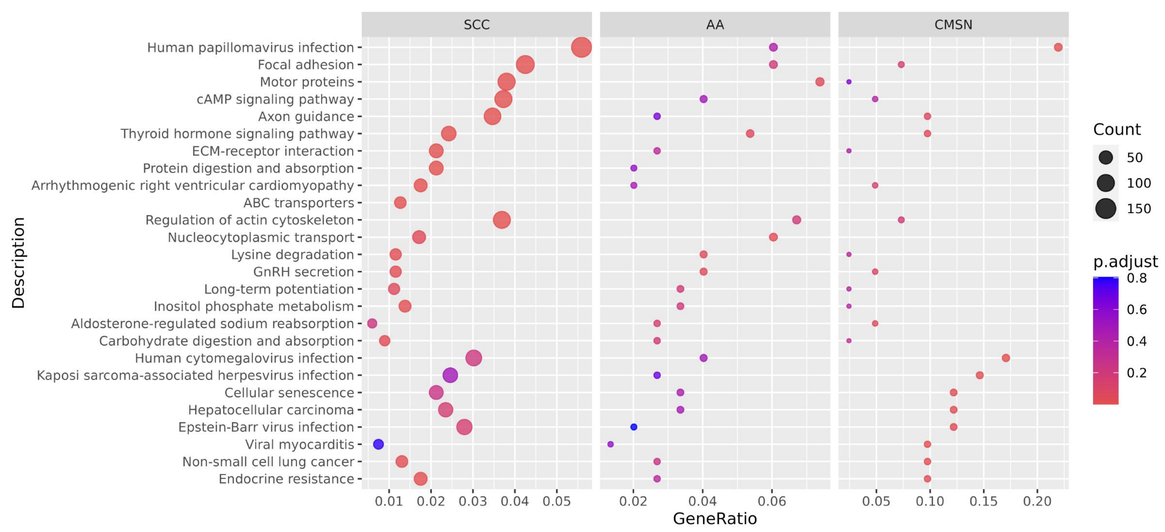
Identification of **shared** and **unique potential target genes** and cancer pathways across three types of cervical cancer—squamous cell carcinoma, adenocarcinoma, and rare variants—associated with HPV16 and HPV18 infections. This study enhances understanding of the genetic differences and the role of high-risk HPV, aiming to improve diagnostic and therapeutic strategies for cervical cancer.

METHODOLOGY

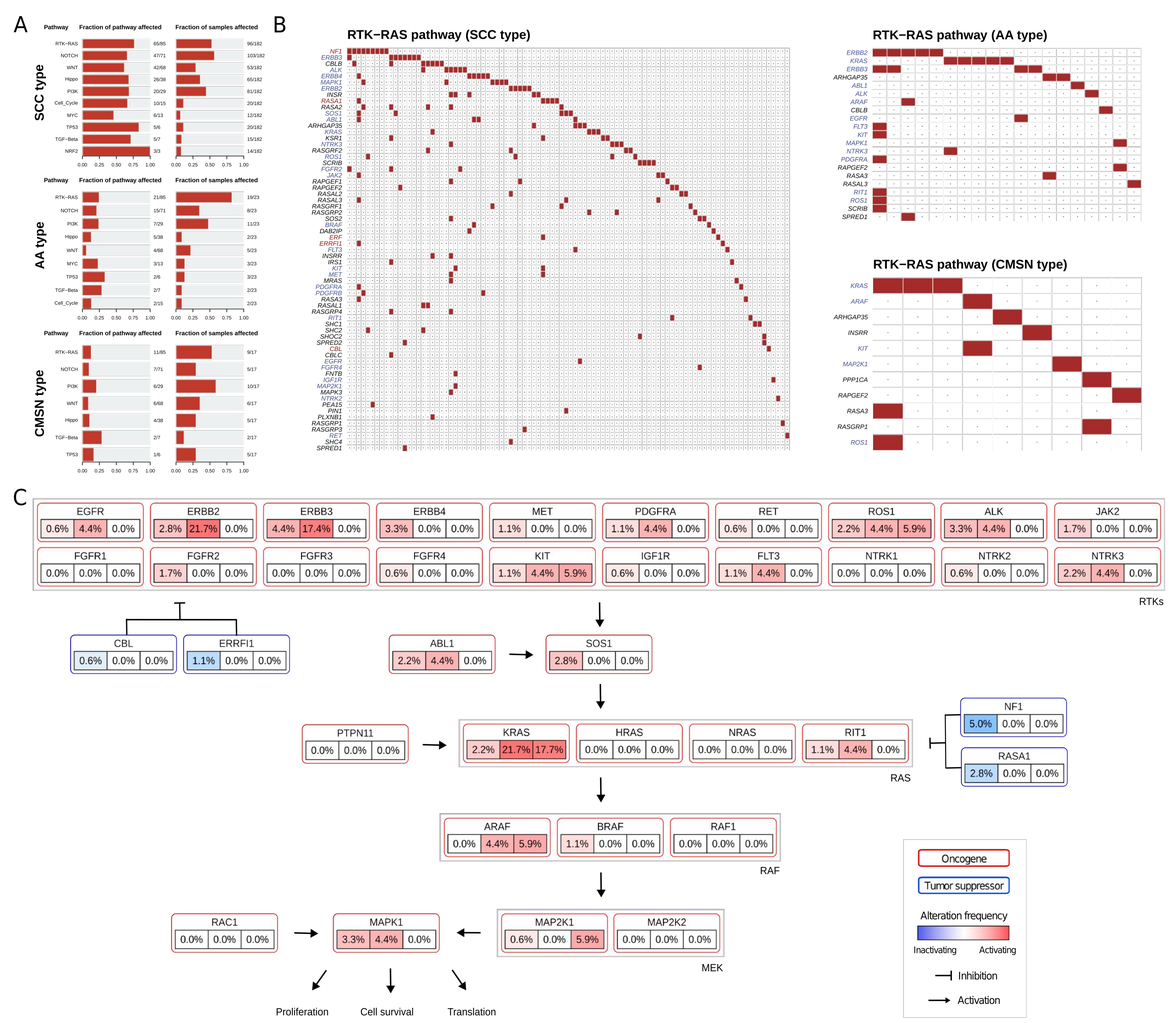


RESULTS

KEGG analysis revealed the "Human papillomavirus infection" pathway in SCC, AA and CMSN, with the "ABC transporters" pathway present in AA and CMSN. Unique pathways included "Nucleocytoplasmic transport" and "Protein digestion and absorption" in CMSN.



The main oncogenic pathways in all three CC types are RTK-RAS, NOTCH, WNT, Hippo, PI3K, TP53, and TGF-β, with RTK-RAS being dominant. Key mutated genes include NF1 (SCC), ERBB2, and KRAS (AA), and KRAS (CMSN). Common affected genes are ROS1, KIT, and KRAS, with KRAS most impacted.



CONCLUSIONS

PIK3CA and KMT2C are linked to the progression of multiple cervical cancer types, with low tumor mutational burden (TMB) suggesting poorer prognosis in SCC and AA. KRAS, KIT, and ROS1 show potential as prognostic biomarkers and therapeutic targets, supporting future clinical research.

PERSPECTIVES

Further research should investigate PIK3CA, KMT2C, KRAS, KIT, and ROS1 for their clinical potential in prognosis and therapy of cervical cancer. Additionally, understanding low TMB in SCC and AA could improve risk stratification and personalized treatments.

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